

# DIGITAL HEALTH PLATFORM

## LEVEL 0 DFD, LEVEL 1 DFD & SYSTEM FLOW DIAGRAM (SFD)

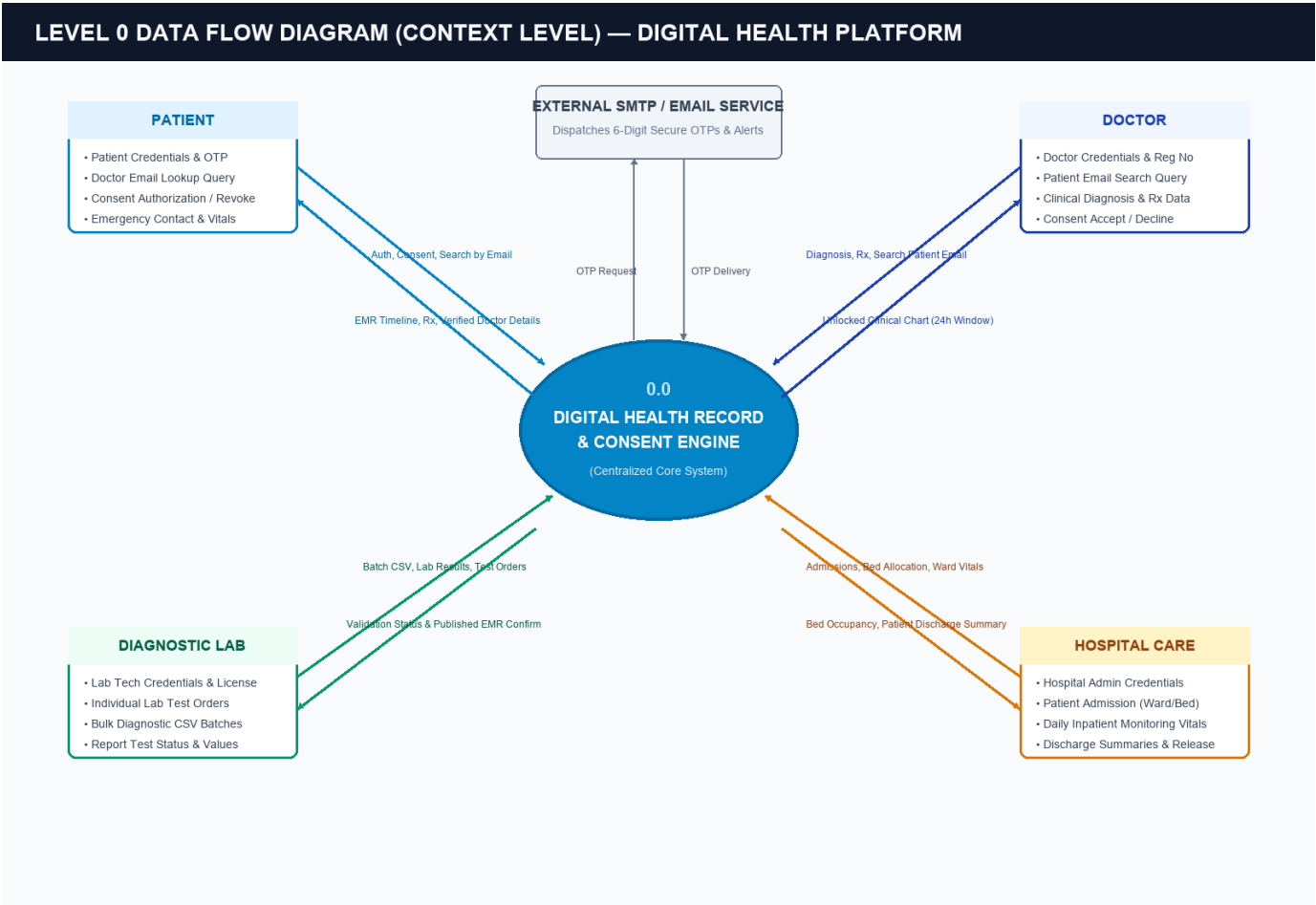
|                    |                                                              |
|--------------------|--------------------------------------------------------------|
| Platform Scope:    | Patient, Doctor, Diagnostic Lab & Hospital Inpatient Modules |
| Security Standard: | HIPAA / ABDM Compliant 24-Hour Clinical Consent Protocol     |
| Discovery Engine:  | Strict Email-Based Verification (Zero Public Directory Leak) |
| Backend Stack:     | Django REST Framework, PostgreSQL / SQLite, JWT + SMTP OTP   |
| Frontend Stack:    | Flutter Web & Mobile Reactive State (Riverpod)               |
| Document Type:     | System Engineering Diagrams & Functional Specifications      |

### Executive Summary & Core Architectural Highlights

This engineering report provides official visual diagrams and architectural models for the Digital Health Platform. The system decouples clinical roles while ensuring that sensitive medical history is accessible only with active, patient-consented time-bounded windows. Zero arbitrary directory listing prevents mass harvesting of doctor or patient accounts, upholding patient rights and HIPAA standards.

# 1. Level 0 Data Flow Diagram (Context Level)

The Level 0 Context Diagram establishes the global boundary of the Digital Health Platform. The central system (Process 0.0) interacts with four primary external entities (Patient, Doctor, Diagnostic Lab, Hospital Care) and an external SMTP notification service.

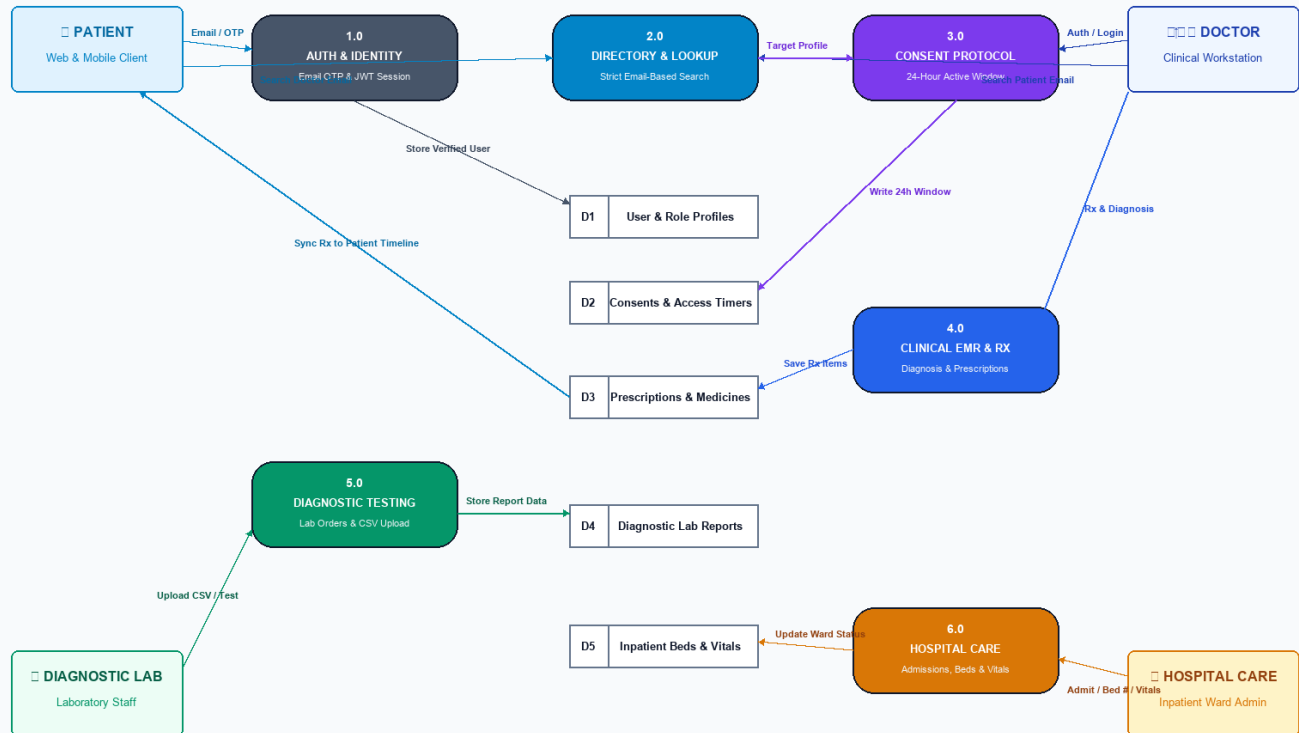


| External Entity | Input Data Flows into System                                          | Output Data Flows to Entity                                                                           |
|-----------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Patient         | Credentials, OTP, Doctor Email Query, Consent Grant/Revoke            | Auth, Consent, Search by Email, EMR Timeline, Rx, Verified Doctor Details, Prescriptions, Lab Reports |
| Doctor          | Doctor Credentials, Patient Email Query, Diagnosis, Meds, Rx, Consent | Diagnosis, Rx, Search Patient Email, Unlocked Clinical EMR Chart (24h Window)                         |
| Diagnostic Lab  | Lab Tech Auth, Single Test Orders, Batch CSV Uploads                  | Batch Parsing Status, Published EMR Record Confirmation                                               |
| Hospital Care   | Admin Auth, Inpatient Admission, Ward/Bed Assignment, Daily Vitals    | Admissions, Bed Allocation, Ward Vitals, Bed Occupancy Dashboard, Discharge Summary                   |

## 2. Level 1 Data Flow Diagram (Subsystem Decomposition)

The Level 1 DFD decomposes the central process into 6 specialized subprocesses and 5 centralized persistent data stores (D1 to D5). Strict email search rules guarantee that no entity can enumerate public clinical records without authorization.

### LEVEL 1 DATA FLOW DIAGRAM — FUNCTIONAL SUBSYSTEMS DECOMPOSITION

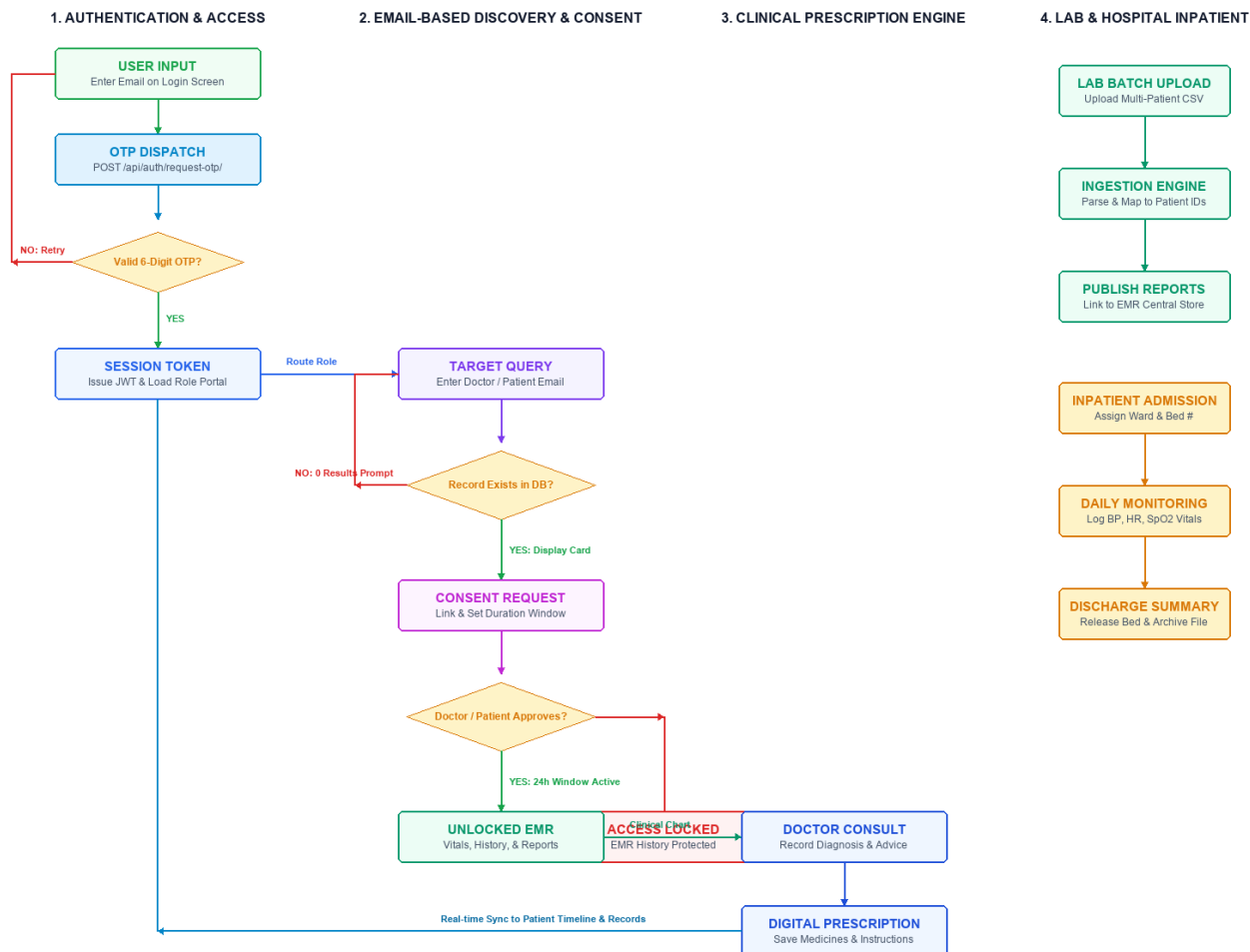


| Process ID | Subsystem Name            | Primary Data Inputs                        | Target Data Store            |
|------------|---------------------------|--------------------------------------------|------------------------------|
| 1.0        | Authentication & Identity | Email, OTP, Role Profile Payload           | D1: Users & Role Profiles    |
| 2.0        | Directory & Lookup        | Target Email Query (Doctor / Patient)      | D1: Read-Only Verification   |
| 3.0        | Consent Protocol          | Requested Duration (24h/7d), Doctor ID     | D2: Consents & Access Timers |
| 4.0        | Clinical EMR & Rx         | Diagnosis, Drug Name, Dosage, Instructions | D3: Prescriptions & Items    |
| 5.0        | Diagnostic Testing        | Lab Order Form, Bulk CSV File              | D4: Diagnostic Lab Reports   |
| 6.0        | Hospital Care             | Admission Code, Ward/Bed #, Daily Vitals   | D5: Inpatient Beds & Vitals  |

### 3. System Flow Diagram (SFD) — Operational & Decision Logic

The System Flow Diagram details step-by-step branching logic, decision diamonds, database validation, and rollback operations across Authentication, Discovery, Consent Locking, and Prescription Delivery.

#### SYSTEM FLOW DIAGRAM (SFD) — COMPLETE WORKFLOW & DECISION LOGIC



| Decision Check       | Evaluation Rule                       | Success Path                        | Failure / Rollback Path                |
|----------------------|---------------------------------------|-------------------------------------|----------------------------------------|
| Valid 6-Digit OTP?   | Match against EmailOTP within 5 mins  | Issue JWT & Redirect to Role Portal | Increment attempts; lock on 3 fails    |
| Record Exists in DB? | Query User by email__iexact           | Render Verified Profile Badge       | Return 0 results; display error prompt |
| Consent Approved?    | Check status='approved' & valid_until | Unlock 24h Clinical History Chart   | Enforce Consent Required Lock Screen   |
| Valid CSV Ingestion? | Validate email, headers, test values  | Store batch & publish to EMR        | Reject batch; display error report     |